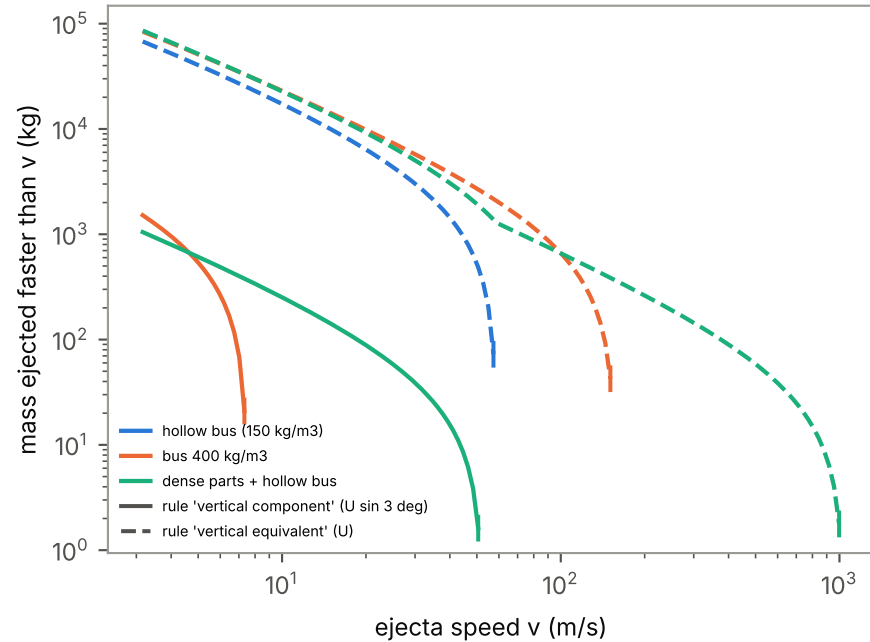
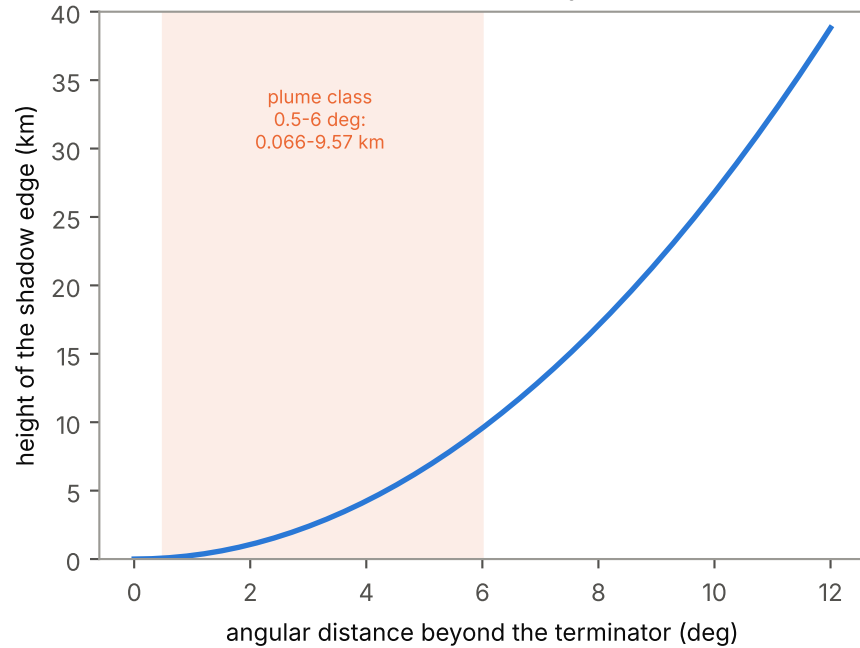


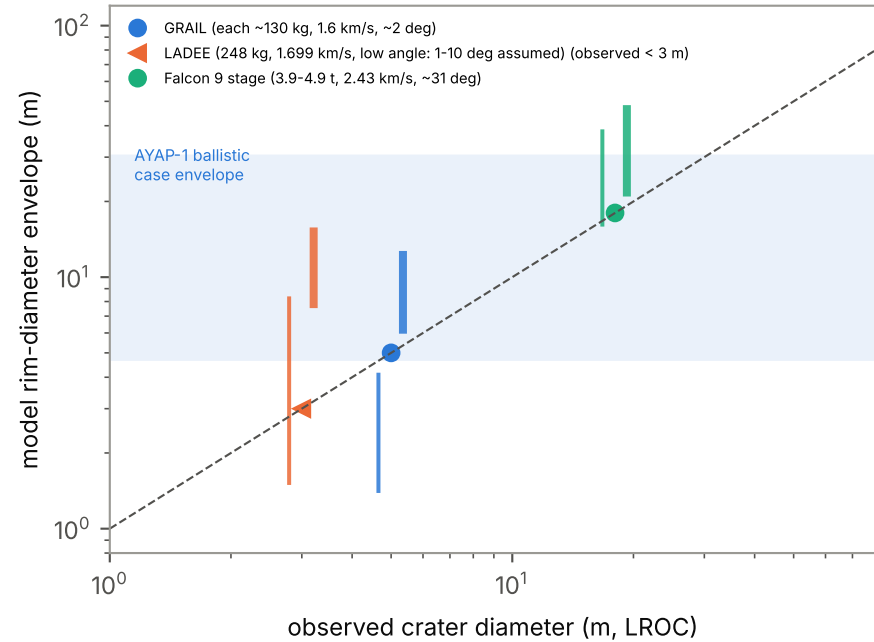
a) Ejecta mass-speed, ballistic case (HH2011, sand);
curves stop at the scaling-domain edge (bar): faster = cannot determine



b) Height ejecta must reach to be sunlit (point Sun, sphere;
the finite solar disc and terrain broaden the edge)



c) Consistency check, no tuning (thin: vertical-component rule;
thick: vertical-equivalent; parameter sets x densities 150-1000)



MODEL; panel c compares with OBSERVED LROC crater sizes (research/precedents.csv); not a validation of either oblique-impact rule